Trigonometry-1

Important Formula (I):

1) Sum/Diff to product Formula :→

$\sin(A+B) = \sin A\cdot\cos B + \cos A\cdot\sin B$

$\sin(A-B) = \sin A\cdot\cos B - \cos A\cdot\sin B$

$\cos(A+B) = \cos A\cdot\cos B - \sin A\cdot\sin B$

$\cos(A-B) = \cos A\cdot\cos B + \sin A\sin B$

$$\tan(A+B) = \frac{\tan A+\tan B}{1-\tan A\cdot\tan B}$$

$$\tan(A-B) = \frac{\tan A-\tan B}{1+\tan A\cdot\tan B}$$

2) Double Angle & Triple Angle Formula :→

(i) $\sin(2A) = 2\sin A\cdot\cos A = \frac{2\tan A}{1+\tan^2 A}$

(ii) $\cos(2A) = 2\cos^2A - 1 = 1 - 2\sin^2A = (\cos^2A - \sin^2A) = \frac{1-\tan^2A}{1+\tan^2A}$

(iii) $\sin 3A = 3\sin A - 4\sin^3A$

(iv) $\cos 3A = 4\cos^3A - 3\cos A$

Ques If $2(\cos\beta - \cos\alpha) + \cos\alpha\cdot\cos\beta = 1$, then find relation(s) b/w $\tan(\alpha/2)$ & $\tan(\beta/2)$ [JA-2017]

Sol^n $2(\cos\beta - \cos\alpha) + \cos\alpha\cdot\cos\beta = 1$

$2\cos\beta + \cos\alpha\cdot\cos\beta = 1 + 2\cos\alpha$

$\cos\beta\,(2+\cos\alpha) = 1 + 2\cos\alpha$

$$\Rightarrow \cos\beta = \frac{1+2\cos\alpha}{2+\cos\alpha}$$

$$\frac{1-\tan^2(\beta/2)}{1+\tan^2(\beta/2)} = \frac{1+2\left[\frac{1-\tan^2(\alpha/2)}{1+\tan^2(\alpha/2)}\right]}{2+\left[\frac{1-\tan^2(\alpha/2)}{1+\tan^2(\alpha/2)}\right]}$$

$$\Rightarrow \frac{1-\tan^2(\beta/2)}{1+\tan^2(\beta/2)} = \frac{3-\tan^2(\alpha/2)}{3+\tan^2(\alpha/2)}$$

$$\frac{1-\tan^2(\beta/2)}{1+\tan^2(\beta/2)} = \frac{1-\left(\frac{\tan^2(\alpha/2)}{3}\right)}{1+\left(\frac{\tan^2(\alpha/2)}{3}\right)}$$

On comparing

$$\tan^2(\beta/2) = \frac{\tan^2(\alpha/2)}{3}$$

$$\Rightarrow \tan(\alpha/2) = \pm\sqrt{3}\tan(\beta/2)$$

So, Relations → ① $\tan(\alpha/2) + \sqrt{3}\tan(\beta/2) = 0$; ② $\tan(\alpha/2) - \sqrt{3}\tan(\beta/2) = 0$ **Ans**

Ques Find value of $\tan3A\cdot\tan2A\cdot\tan A$

Sol^n $\because 3A = 2A + A$

$\tan(3A) = \tan(2A+A)$

$$\tan(3A) = \frac{\tan2A+\tan A}{1-\tan A\cdot\tan2A}$$

$\Rightarrow \tan3A - \tan3A\cdot\tan2A\cdot\tan A = \tan2A + \tan A$

$\Rightarrow \boxed{\tan3A\cdot\tan2A\cdot\tan A = \tan3A - \tan2A - \tan A}$ **Ans**

Ques If $5(\tan^2x - \cos^2x) = \frac{2\cos x + 9}{\cos(x)}$, find $\cos(4x)$.

Sol^n $5\left(\frac{\sin^2x}{\cos^2x} - \cos^2x\right) = 2(2\cos^2x - 1) + 9$

Let, $\cos^2x = t$ and put $\sin^2x = 1-\cos^2x$

$\Rightarrow 5\left(\frac{1-t}{t} - t\right) = 2(2t-1) + 9$

$\Rightarrow 9t^2 + 12t - 5 = 0$ → $t = 1/3$ (✓), → $t = -5/3$ (✗)

(Because $\cos^2x$ can't be −ve)

Now, $\cos4x = 2\cos^2(2x) - 1$

$= 2[2\cos^2x - 1]^2 - 1$

$= 2[2(1/3) - 1]^2 - 1$

$\Rightarrow \boxed{\cos4x = -7/9}$ **Ans**

Ques If $\frac{\sin x}{\sin y} = 3$ and $\frac{\cos x}{\cos y} = \frac{1}{2}$, then find $\frac{\sin2x}{\sin2y} + \frac{\cos2x}{\cos2y}$.

Sol^n $\frac{\sin2x}{\sin2y} + \frac{\cos2x}{\cos2y} = \frac{2\sin x\cos x}{2\sin y\cos y} + \frac{2\cos^2x - 1}{2\cos^2y - 1}$

$= (3)\left(\frac{1}{2}\right) + \frac{2\cos^2x - 1}{2\cos^2y - 1}$

$\because \sin x = 3\sin y$; $\cos x = \frac{\cos y}{2}$

$\sin^2x + \cos^2x = 1$

$9\sin^2y + \frac{\cos^2y}{4} = 1$

$9(1-\cos^2y) + \frac{\cos^2y}{4} = 1$

$\Rightarrow \boxed{\cos^2y = 32/35}$

$\cos^2x = \frac{\cos^2y}{(2)^2}$

$\Rightarrow \boxed{\cos^2x = 8/35}$

So,

Expression $= \frac{3}{2} + \frac{2(8/35) - 1}{2(32/35) - 1} \Rightarrow \frac{49}{58}$ **Ans**

Ques Simplify $(4\cos^29° - 3)\cdot(4\cos^227° - 3) = \tan9°$

Sol^n $\because \cos3\theta = 4\cos^3\theta - 3\cos\theta$

$\Rightarrow \left(\frac{\cos3\theta}{\cos\theta} = 4\cos^2\theta - 3\right)$

L.H.S = $\left(\frac{\cos27°}{\cos9°}\right)\left(\frac{\cos81°}{\cos27°}\right) \Rightarrow \frac{\cos81°}{\cos9°}$

$= \frac{\sin9°}{\cos9°} \Rightarrow \tan9°$ **Ans**

Ques If $\cos(\alpha+\beta) = \frac{4}{5}$ and $\sin(\alpha-\beta) = 5/13$, where $0 \le \alpha,\beta \le \pi/4$, then find $\tan(2\alpha)$.

Sol^n $\tan(2\alpha) = \tan[(\alpha+\beta) + (\alpha-\beta)]$

$$\tan(2\alpha) = \frac{\tan(\alpha+\beta) + \tan(\alpha-\beta)}{1 - \tan(\alpha+\beta)\cdot\tan(\alpha-\beta)}$$

$\cos(\alpha+\beta) = \frac{4}{5}$; $\sin(\alpha-\beta) = 5/13$

$B = 4$, $H = 5$ $\Rightarrow P = \sqrt{25-16} = 3$

$\boxed{\tan(\alpha+\beta) = 3/4}$

$P = 5$, $H = 13$ $\Rightarrow B = \sqrt{169-25} = 12$

$\boxed{\tan(\alpha-\beta) = 5/12}$

So, $\tan(2\alpha) = \frac{3/4 + 5/12}{1 - (3/4)(5/12)} = \frac{56}{33}$ **Ans**

Ques α & β are roots of $(k+1)\tan^2x - \sqrt{2}\lambda\tan x = (1-k)$ where $k \ne -1$ & $k \in R$. Then find λ if $\tan^2(\alpha+\beta) = 50$.

Sol^n $(k+1)\tan^2x - \sqrt{2}\lambda\tan x = (1-k)$ → $\tan\alpha$, → $\tan\beta$

$\tan\alpha + \tan\beta = \frac{\sqrt{2}\lambda}{k+1}$; $\tan\alpha\cdot\tan\beta = \frac{k-1}{k+1}$

$$\left(\tan(\alpha+\beta)\right)^2 = \left(\frac{\tan\alpha+\tan\beta}{1-\tan\alpha\cdot\tan\beta}\right)^2$$

So $= \left(\frac{\sqrt{2}\lambda}{2}\right)^2 \Rightarrow \boxed{\lambda = \pm10}$ **Ans**

Ques Let a, b, c be 3 non-zero real no. such that the eqn $\sqrt{3}a\cos x + 2b\sin x = c$, $x \in [-\pi/2, \pi/2]$ has two distinct roots α and β with $\alpha+\beta = \pi/3$. Find b/a.

Sol^n Since α & β are roots of eqn

$\boxed{\sqrt{3}a\cos\alpha + 2b\sin\alpha = c}$ — ①

$\sqrt{3}a\cos\beta + 2b\sin\beta = c$

$\sqrt{3}a\cos(13-\alpha) + 2b\sin(13-\alpha) = c$

$$\boxed{\sqrt{3}a\left[\frac{\cos\alpha}{2} + \frac{\sqrt{3}\sin\alpha}{2}\right] + 2b\left[\frac{\sqrt{3}\cos\alpha}{2} - \frac{\sin\alpha}{2}\right] = c}$$ — ②

By ① & ②, we can equate L.H.S of both eqn & on comparing coefficients, we get

$\cos\alpha\left(\frac{\sqrt{3}a}{2} + b\sqrt{3}\right) + \sin\alpha\left(\frac{3a}{2} - b\right) = (\cos\alpha)(\sqrt{3}a) + (\sin\alpha)(2b)$

$\frac{\sqrt{3}a}{2} + b\sqrt{3} = \sqrt{3}a \Rightarrow b\sqrt{3} = \frac{\sqrt{3}}{2}a$

$\Rightarrow \boxed{\frac{b}{a} = \frac{1}{2}}$ **Ans**

Important Formulas (II):

1) $\tan\alpha\cdot\tan(60°-\alpha)\cdot\tan(60°+\alpha) = \tan3\alpha$

2) $\sin\alpha\cdot\sin(60°-\alpha)\cdot\sin(60°+\alpha) = \frac{\sin3\alpha}{4}$

3) $\cos\alpha\cdot\cos(60°-\alpha)\cdot\cos(60°+\alpha) = \frac{\cos3\alpha}{4}$

NOTE: $\sin 15^\circ = \frac{\sqrt{3}-1}{2\sqrt{2}}$ | $\cos 15^\circ = \frac{\sqrt{3}+1}{2\sqrt{2}}$

Eg; Find $\tan 6^\circ \cdot \tan 66^\circ \cdot \tan 42^\circ \cdot \tan 78^\circ$.

Sol^n $\frac{\tan 6^\circ \cdot \tan 66^\circ \tan 54^\circ}{\tan 54^\circ} \quad \frac{\tan 42^\circ \cdot \tan 78^\circ \tan 18^\circ}{\tan 18^\circ}$

$= \frac{\tan 18^\circ}{\tan 54^\circ} \times \frac{\tan 54^\circ}{\tan 18^\circ} \Rightarrow 1$ Ans

Ques Find $\sin 10^\circ \cdot \sin 30^\circ \cdot \sin 50^\circ \cdot \sin 70^\circ$.

Sol^n $= (\sin 10^\circ \cdot \sin 50^\circ \cdot \sin 70^\circ) \sin 30^\circ$

$= \frac{\sin 30^\circ}{4} \times \sin 30^\circ \Rightarrow \frac{1}{16}$ Ans

Important Observation Ques

Ques If $\sin\alpha + \sin\beta + \sin\gamma = 0$ and $\cos\alpha + \cos\beta + \cos\gamma = 0$, Find $\cos(\alpha-\beta) + \cos(\beta-\gamma) + \cos(\gamma-\alpha)$ (Let R)

Sol^n $\cos(\alpha-\beta) = \cos\alpha \cdot \cos\beta + \sin\alpha \cdot \sin\beta$

$\cos(\beta-\gamma) = \cos\beta \cdot \cos\gamma + \sin\beta \cdot \sin\gamma$

$\cos(\gamma-\alpha) = \cos\gamma \cdot \cos\alpha + \sin\gamma + \sin\alpha$

Now, we get two terms in product when we square three terms like $(a+b+c)^2$.

So,

$(\sin\alpha + \sin\beta + \sin\gamma)^2 = \sin^2\alpha + \sin^2\beta + \sin^2\gamma + 2(\sin\alpha \cdot \sin\beta + \sin\beta \cdot \sin\gamma + \sin\alpha \cdot \sin\gamma)$ — ①

$(\cos\alpha + \cos\beta + \cos\gamma)^2 = \cos^2\alpha + \cos^2\beta + \cos^2\alpha + 2(\cos\alpha \cdot \cos\beta + \cos\beta \cdot \cos\gamma + \cos\alpha \cdot \cos\gamma)$ — ②

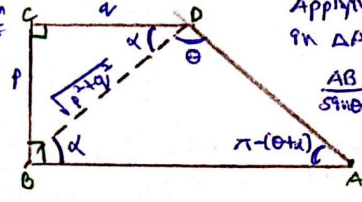
① + ② ⇒

$0 = (\sin^2\alpha + \cos^2\alpha) + (\sin^2\beta + \cos^2\beta) + (\sin^2\gamma + \cos^2\gamma) + 2(R)$

$\Rightarrow \boxed{R = -3/2}$ Ans

Ques ABCD is a trapezium such that AB and CD are parallel and BC ⊥ CD. If $\angle ADB = \theta$, BC = p and CD = q then find AB.

Sol^n



Applying sine rule in ΔABD.

$\frac{AB}{\sin\theta} = \frac{AD}{\sin\alpha} = \frac{\sqrt{p^2+q^2}}{\sin(\theta+\alpha)}$ [∵ $\sin(\pi-x) = \sin x$]

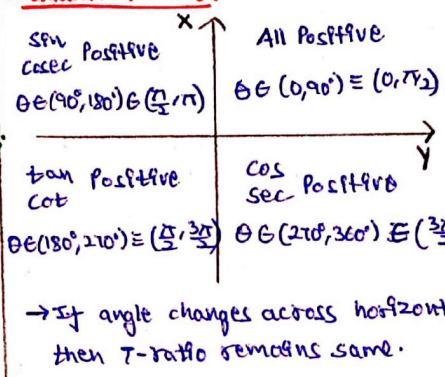
Now,

$AB = \frac{\sqrt{p^2+q^2}}{\sin\theta \cdot \cos\alpha + \sin\alpha \cdot \cos\theta} \cdot \sin\theta$

$\Rightarrow AB = \frac{\sqrt{p^2+q^2} \cdot \sin\theta}{\sin\theta \cdot \frac{q}{\sqrt{p^2+q^2}} + \cos\theta \cdot \frac{p}{\sqrt{p^2+q^2}}}$

$\boxed{AB = \frac{\sin\theta (p^2+q^2)}{q\sin\theta + p\cos\theta}}$ Ans

Quadrant Rule:



→ If angle changes across horizontal (x-axis) then T-ratio remains same.

→ If angle changes across vertical (y-axis) then T-ratios gets changed as

sin ↔ cos | tan ↔ cot | sec ↔ cosec

→ Sign of final T-ratio is decided by the sign of original T-ratio in the resulting quadrant.

Eg: $\sin 135^\circ = \sin(180^\circ - 45^\circ) = \sin 45^\circ$

$\cos 285^\circ = \cos(270^\circ + 15^\circ) = \sin 15^\circ$

$\tan 335^\circ = \tan(360^\circ - 25^\circ) = -\tan 25^\circ$

$\sec\frac{13\pi}{12} = \sec\left(\pi + \frac{\pi}{12}\right) = -\sec\left(\frac{\pi}{12}\right)$

Ques Let $\frac{\sqrt{2}\sin\alpha}{\sqrt{1+\cos 2\alpha}} = \frac{1}{7}$ and $\sqrt{\frac{1-\cos 2\beta}{2}} = \frac{1}{\sqrt{10}}$ where $\alpha, \beta \in (0, \pi/2)$ then $\tan(\alpha+2\beta)$ is equal to ____.

Sol^n $\frac{\sqrt{2}\sin\alpha}{\sqrt{1+\cos 2\alpha}} = \frac{\sqrt{2}\sin\alpha}{\sqrt{1+2\cos^2\alpha - 1}} = \tan\alpha = \frac{1}{7}$

$\sqrt{\frac{1-\cos 2\beta}{2}} = \sqrt{\frac{1-(1-2\sin^2\beta)}{2}} \Rightarrow \sin\beta = \frac{1}{\sqrt{10}}$

$\cos\beta = \frac{3}{\sqrt{10}}$

$\Rightarrow \tan\beta = \frac{1}{3}$

Now, $\tan 2\beta = \frac{2\tan\beta}{1-\tan^2\beta}$

$\Rightarrow \tan 2\beta = \frac{2(1/3)}{1-1/9} \Rightarrow \frac{3}{4}$

So, $\tan(\alpha+2\beta) = \frac{\tan\alpha + \tan 2\beta}{1 - \tan\alpha \cdot \tan 2\beta}$

$= \frac{1/7 + 3/4}{1 - (\frac{1}{7})(\frac{3}{4})} \Rightarrow 1$ Ans

Important Formula (Part III):

1) $\sin A + \sin B = 2\sin\left(\frac{A+B}{2}\right) \cdot \cos\left(\frac{A-B}{2}\right)$

2) $\sin A - \sin B = 2\cos\left(\frac{A+B}{2}\right) \cdot \sin\left(\frac{A-B}{2}\right)$

3) $\cos A + \cos B = 2\cos\left(\frac{A+B}{2}\right) \cdot \cos\left(\frac{A-B}{2}\right)$

4) $\cos A - \cos B = -2\sin\left(\frac{A+B}{2}\right) \cdot \sin\left(\frac{A-B}{2}\right)$

Ques If $0 < \alpha, \beta < \pi$ and $\cos\alpha + \cos\beta - \cos(\alpha+\beta) = 3/2$, find α and β.

Sol^n $\cos\alpha + \cos\beta - \cos(\alpha+\beta) = 3/2$

$2\cos\left(\frac{\alpha+\beta}{2}\right) \cdot \cos\left(\frac{\alpha-\beta}{2}\right) - \left[2\cos^2\left(\frac{\alpha+\beta}{2}\right) - 1\right] = \frac{3}{2}$

$4\cos\left(\frac{\alpha+\beta}{2}\right) \cdot \cos\left(\frac{\alpha-\beta}{2}\right) - 4\cos^2\left(\frac{\alpha+\beta}{2}\right) + 2 = 3$

$4\cos^2\left(\frac{\alpha+\beta}{2}\right) - 2\cos\left(\frac{\alpha+\beta}{2}\right) \cos\left(\frac{\alpha-\beta}{2}\right) = -1$

On completing square ⇒

$\left[2\cos\left(\frac{\alpha+\beta}{2}\right) - \cos\left(\frac{\alpha-\beta}{2}\right)\right]^2 = \left[\cos\left(\frac{\alpha-\beta}{2}\right)\right]^2 - 1$

$\Rightarrow \left[2\cos\left(\frac{\alpha+\beta}{2}\right) - \cos\left(\frac{\alpha-\beta}{2}\right)\right]^2 + \sin^2\left(\frac{\alpha-\beta}{2}\right) = 0$

Concept: Whenever $A^2 + B^2 = 0$ $(A, B \in R)$ then $A = B = 0$

So,

$2\cos\left(\frac{\alpha+\beta}{2}\right) = \cos\left(\frac{\alpha-\beta}{2}\right)$ | $\sin^2\left(\frac{\alpha-\beta}{2}\right) = 0$

$2\cos\beta - \cos(0) = 0$ | $\Rightarrow \frac{\alpha-\beta}{2} = 0$

$\cos\beta = 1/2$ | $\alpha = \beta$

$\alpha = \beta = \pi/3$ Ans

Gyan-1: 1) We can write $\sin\theta \cdot \cos\theta = \frac{\sin 2\theta}{2}$

2) $|\sin\theta + \cos\theta| = \sqrt{1 + \sin 2\theta}$

Ques If $0 < \beta < \pi/2$ and $(\sin\beta + \cos\beta + \tan\beta + \cot\beta + \sec\beta + \operatorname{cosec}\beta = 7)$ then find eqn whose root is $\sin 2\beta$.

Sol^n $(\sin\beta + \cos\beta) + \frac{\sin\beta}{\cos\beta} + \frac{\cos\beta}{\sin\beta} + \frac{1}{\sin\beta} + \frac{1}{\cos\beta} = 7$

$\Rightarrow \sqrt{1+\sin 2\beta} + \frac{1}{\sin\beta \cdot \cos\beta} + \frac{\sin\beta + \cos\beta}{\sin\beta \cdot \cos\beta} = 7$

$\sqrt{1+\sin 2\beta} + \frac{2}{\sin 2\beta} + \frac{2\sqrt{1+\sin 2\beta}}{\sin 2\beta} = 7$

[Let, $\sin 2\beta = x$ and $\sin 2\beta \neq 0 \Rightarrow x \neq 0$]

So $\sqrt{1+x} + \frac{2}{x} + \frac{2\sqrt{1+x}}{x} = 7$

$\sqrt{1+x}\left(1 + \frac{2}{x}\right) = \frac{7x-2}{x}$

$(\sqrt{1+x})(x+2) = (7x-2)$

$(1+x)(x^2+4x+4) = 49x^2 + 4 - 28x$

$\Rightarrow x^3 - 44x^2 + 36x = 0$

$x(x^2 - 44x + 36) = 0$

$x = 0$ (x) | $x^2 - 44x + 36 = 0$ (✓)

Ans

Product of cosine's angles in GP:

$\cos\alpha \cdot \cos 2\alpha \cdot \cos 2^2\alpha \cdot \cos 2^3\alpha \ldots \cos 2^{n-1}\alpha = \frac{\sin 2^n\alpha}{2^n \cdot \sin\alpha}$

4 Always use this formula whenever cosine series appears in multiplication and angles get doubled with each term.

Ques The value of $\cos\frac{\pi}{2^1} \cdot \cos\frac{\pi}{2^2} \ldots \cos\frac{\pi}{2^{10}} \cdot \sin\frac{\pi}{2^{10}}$ is ____.

Trigonometry-2

Sol^n $\left(\cos\frac{\pi}{2^{10}}\cdot\cos\frac{\pi}{2^9}\cdots\cos\frac{\pi}{2^3}\cdot\cos\frac{\pi}{2^2}\right)\sin\frac{\pi}{2^{10}}$

$n-1=8 \Rightarrow n=9$

So,

$$\frac{\sin\left[2^9\left(\frac{\pi}{2^{10}}\right)\right]}{2^9\sin\left(\frac{\pi}{2^{10}}\right)}\cdot\sin\left(\frac{\pi}{2^{10}}\right)$$

$\Rightarrow \frac{\sin(\pi/2)}{2^9} \Rightarrow \frac{1}{512}$ Ans

Gyan-2: If $A+B=90°$, then

$$\sin A\cdot\sin B=\frac{\sin 2A}{2} \text{ or } \frac{\sin 2B}{2}$$

(Here, A → Smaller Angle)

Eg: $\frac{\sin\frac{\pi}{16}}{1}\cdot\frac{\sin\frac{3\pi}{16}}{1}\cdot\frac{\sin\frac{5\pi}{16}}{2}\cdot\frac{\sin\frac{7\pi}{16}}{2}$

Sol^n

$\Rightarrow \frac{\sin\left(\frac{\pi}{8}\right)}{2}\cdot\frac{\sin\left(\frac{3\pi}{8}\right)}{2}$ $\left(\frac{\pi}{8}+\frac{3\pi}{8}=\frac{\pi}{2}\right)$

$= \frac{\sin(\pi/4)}{8} \Rightarrow \frac{1}{8\sqrt{2}}$ Ans

Gyan-3: $\frac{1-\cos 2A}{\sin 2A}=\tan A$

$\tan 22.5°=\sqrt{2}-1$

$\cot 22.5°=\sqrt{2}+1$

$(\pi/8 = 22.5°)$

Gyan-4: $\frac{1+\cos 2A}{\sin 2A}=\cot A$

$\cot 7.5°=\sqrt{2}+\sqrt{3}+\sqrt{4}+\sqrt{6}$

Gyan-5: Somewhere, if it is given

$a\sin\alpha+b\cos\alpha=R$

Then to solve the eqn, multiply divide by $\sqrt{a^2+b^2}$

Eg: Find value of $\sqrt{3}\operatorname{cosec}20°-\sec 20°$.

Sol^n $\frac{\sqrt{3}}{\sin 20°}-\frac{1}{\cos 20°}$

$= \frac{\sqrt{3}\cos 20°-\sin 20°}{\frac{1}{2}\sin 20°\cdot\cos 20°}$ $\left[\sqrt{(\sqrt{3})^2+(1)^2}=\sqrt{4}=2\right]$

$= \frac{2\left(\frac{\sqrt{3}}{2}\cdot\cos 20°-\frac{1}{2}\sin 20°\right)}{\frac{1}{2}\sin 40°}$

$= \frac{4}{\sin 40°}(\cos 50°) \Rightarrow \frac{4}{\sin 40°}\cdot\sin 40°$

$= 4$ Ans

Gyan-6: If $A+B=90°$, then

$$\sin^2 A+\sin^2 B=1$$

Eg: Find value of $\sin^2 5°+\sin^2 10°\cdots+\sin^2 90°$

Sol^n $\sin^2 90°+\left(\sin^2 5°+\sin^2 85°\right)+\left(\sin^2 10°+\sin^2 80°\right)\cdots\left(\sin^2 40°+\sin^2 50°\right)+\sin^2 45°$

$1 \quad\quad 1 \quad\quad\quad 1 \quad\quad\quad\quad 1$

So, $= 1+8+\left(\frac{1}{\sqrt{2}}\right)^2 \Rightarrow \frac{19}{2}$ Ans

Gyan-1: If we are given $\sin^4 A+\cos^4 A$ then,

$\sin^4 A+\cos^4 A=\left(\sin^2 A+\cos^2 A\right)^2-2\sin^2 A\cdot\cos^2 A$

$= (1)^2-2\left(\frac{\sin 2A}{2}\right)^2$

$\Rightarrow \left[\sin^4 A+\cos^4 A=1-\frac{\sin^2 2A}{2}\right]$

If $A+B=90°$, then

$$(\sin A)^4+(\sin B)^4=1-\frac{\sin^2(2A)}{2}$$

(A → Smaller Angle)

Eg: Find value of $\sin^4\theta+\sin^4 3\theta+\sin^4 5\theta+\sin^4 7\theta$ where $\theta=\pi/16$.

Sol^n $\left(\sin^4\theta+\sin^4 7\theta\right)+\left(\sin^4 5\theta+\sin^4 3\theta\right)$

$= 1-\frac{\sin^2 2\theta}{2}+1-\frac{\sin^2 6\theta}{2}$

$= 2-\frac{1}{2}\left(\sin^2 2\theta+\sin^2 6\theta\right)$

$= 2-\frac{1}{2}(1) \Rightarrow \frac{3}{2}$ Ans

Important Formulae (IV):

1) $\sin(A+B)\cdot\sin(A-B)=\sin^2 A-\sin^2 B$
$=\cos^2 B-\cos^2 A$

2) $\cos(A+B)\cdot\cos(A-B)=\cos^2 A-\sin^2 B$
$=\cos^2 B-\sin^2 A$

Sine-Cosine Series where Angles in AP:

1) $\cos A+\cos(A+B)+\cos(A+2B)\cdots$
$\cdots+\cos(A+(n-1)B)=\cos\left(A+\frac{(n-1)B}{2}\right)\times\frac{\sin\left(\frac{nB}{2}\right)}{\sin\left(\frac{B}{2}\right)}$

2) $\sin A+\sin(A+B)+\sin(A+2B)\cdots$
$\cdots+\sin(A+(n-1)B)=\sin\left(A+\frac{(n-1)B}{2}\right)\cdot\frac{\sin\left(\frac{nB}{2}\right)}{\sin\left(\frac{B}{2}\right)}$

Eg: $\cos\left(\frac{\pi}{2n+1}\right)+\cos\left(\frac{3\pi}{2n+1}\right)+\cos\left(\frac{5\pi}{2n+1}\right)+\cdots$
$\cdots$ n terms.

Sol^n $= \cos\left(\frac{\pi}{2n+1}+\frac{(n-1)\pi}{2n+1}\right)\cdot\frac{\sin\left(\frac{n\pi}{2n+1}\right)}{\sin\left(\frac{\pi}{2n+1}\right)}$

$= \frac{\cos\left(\frac{n\pi}{2n+1}\right)\cdot\sin\left(\frac{n\pi}{2n+1}\right)}{\sin\left(\frac{\pi}{2n+1}\right)} = \frac{\sin\left(\frac{2n\pi}{2n+1}\right)}{2\sin\left(\frac{\pi}{2n+1}\right)}$

Now, whenever T-ratio is in division, we should always first check for following manipulation.

$\therefore \sin(\pi-\theta)=\sin\theta$

If $\theta=\frac{\pi}{2n+1} \Rightarrow \sin\left(\pi-\frac{\pi}{2n+1}\right)=\sin\left(\frac{\pi}{2n+1}\right)$

So, $\frac{1}{2}\left[\frac{\sin\left(\frac{2n\pi}{2n+1}\right)}{\sin\left(\frac{\pi}{2n+1}\right)}\right]=\frac{1}{2}$ Ans

Telescoping Method:

Working Rule ⇒ Always try to make sin-sin or cos-cos in denominator and proceed as following:

1) $\frac{\sin(A-B)}{\sin A\cdot\sin B}=\frac{\sin A\cdot\cos B-\cos A\cdot\sin B}{\sin A\cdot\sin B}$

$=\cot B-\cot A$

2) $\frac{\sin(A-B)}{\cos A\cdot\cos B}=\frac{\sin A\cdot\cos B-\cos A\cdot\sin B}{\cos A\cdot\cos B}$

$=(\tan A-\tan B)$

→ If $\sin^3\theta$ is given, then use triple angle formula.

$$\sin^3 A=\frac{1}{4}\left(3\sin A-\sin 3A\right)$$

Ques $\tan\frac{x}{2}\cdot\sec x+\tan\frac{x}{2^2}\cdot\sec\frac{x}{2}+\cdots$
$\cdots$ + upto n terms

Sol^n $T_1=\frac{\sin\frac{x}{2}}{\cos\frac{x}{2}}\cdot\frac{1}{\cos x} \Rightarrow \frac{\sin(x-x/2)}{\cos\frac{x}{2}\cdot\cos x}=\tan x-\tan\frac{x}{2}$

$\Rightarrow$ $T_1=\tan x-\tan\frac{x}{2}$

$T_2=\tan\frac{x}{2}-\tan\frac{x}{2^2}$

$T_3=\tan\frac{x}{2^2}-\tan\frac{x}{2^3}$

$\vdots$

$T_n=\tan\frac{x}{2^{n-1}}-\tan\frac{x}{2^n}$

$\left(\tan x-\tan\frac{x}{2^n}\right)$ Ans

Ques $\operatorname{cosec}x+\operatorname{cosec}2x+\operatorname{cosec}4x\cdots$ upto n terms

Sol^n $T_1=\frac{1}{\sin x}\times\frac{\sin x/2}{\sin x/2}$ ← multiplied & divided by $\left(\frac{x}{2}\right)$ bcz angles are ↑ing by a factor of 2

$T_1=\cot B-\cot A \Rightarrow \cot\left(\frac{x}{2}\right)-\cot x$

$T_2=\cot x-\cot 2x$

$\vdots$

$T_n=\cot 2^{n-2}x-\cot 2^{n-1}x$

$\Rightarrow \boxed{\cot\left(\frac{x}{2}\right)-\cot\left(2^{n-1}x\right)}$ Ans

Ques $\sum_{r=1}^{n}3^{r-1}\left(\sin\frac{x}{3^r}\right)^3=$ ____

Sol^n $T_1 = 3^0 \cdot \left(\sin\frac{x}{3}\right)^3 = \frac{1}{4}\left(3\sin\frac{x}{3} - \sin x\right)$

$T_2 = 3^1 \left(\sin\frac{x}{3^2}\right)^3 = \frac{1}{4}\left(3^2\sin\frac{x}{3^2} - 3^1\sin\frac{x}{3^1}\right)$

$\vdots$

$T_n = 3^{n-1}\left(\sin\frac{x}{3^n}\right)^3 = \frac{1}{4}\left(3^n\sin\frac{x}{3^n} - 3^{n-1}\sin\frac{x}{3^{n-1}}\right)$

$$\text{Sum} = \left(\frac{3^n \cdot \sin\frac{x}{3^n} - \sin x}{4}\right) \text{Ans}$$

Ques For $0 < \theta < \pi/2$, the solution(s) of

$$\sum_{m=1}^{6} \operatorname{cosec}\left(\theta + \frac{(m-1)\pi}{4}\right) \cdot \operatorname{cosec}\left(\theta + \frac{m\pi}{4}\right) = 4\sqrt{2} \text{ is (are)}$$

Sol^n
$$T_m = \frac{\sin\left[\left(\theta + \frac{m\pi}{4}\right) - \left(\theta + \frac{(m-1)\pi}{4}\right)\right]}{\sin\left(\theta + \frac{(m-1)\pi}{4}\right) \cdot \sin\left(\theta + \frac{m\pi}{4}\right)} \times \frac{1}{R}$$

(R = numerator)

Now,
$$R = \sin(\pi/4) \Rightarrow 1/\sqrt{2}$$

So,
$$T_m = \frac{\sin\left[\left(\theta + \frac{m\pi}{4}\right) - \left(\theta + \frac{(m-1)\pi}{4}\right)\right]}{\sin\left(\theta + \frac{(m-1)\pi}{4}\right) \cdot \sin\left(\theta + \frac{m\pi}{4}\right)} \times \sqrt{2}$$

$$\Rightarrow T_m = \sqrt{2}\left[\cot\left(\frac{(m-1)\pi}{4} + \theta\right) - \cot\left(\frac{m\pi}{4} + \theta\right)\right]$$

Now,

$T_1 = \sqrt{2}\left[\cot\theta - \cot(\theta + \pi/4)\right]$

$T_2 = \sqrt{2}\left[\cot\left(\theta + \frac{\pi}{4}\right) - \cot\left(\theta + \frac{2\pi}{4}\right)\right]$

$\vdots$

$T_6 = \sqrt{2}\left[\cot\left(\theta + \frac{5\pi}{4}\right) - \cot\left(\theta + \frac{6\pi}{4}\right)\right]$

$$\text{Sum} = \left[\cot\theta - \cot\left(\theta + \frac{6\pi}{4}\right)\right]\sqrt{2} = 4\sqrt{2}$$

$$\Rightarrow \cot\theta - \cot\left(\frac{3\pi}{2} + \theta\right) = 4$$

$$\cot\theta + \tan\theta = 4$$

$$\frac{\cos\theta}{\sin\theta} + \frac{\sin\theta}{\cos\theta} = 4$$

$$\frac{2}{\sin 2\theta} = 4 \cdot 2 \Rightarrow \sin 2\theta = \frac{1}{2}$$

$$\Rightarrow 2\theta = \frac{\pi}{6}, \frac{5\pi}{6} \Rightarrow \boxed{\theta = \frac{\pi}{12}, \frac{5\pi}{12}} \text{ Ans}$$

Conditional Identities :

Condition: $A+B+C = \pi$ and angles of Δ.

Then, following identities are valid

1) $\sin(2A) + \sin(2B) + \sin(2C) = 4\sin A \cdot \sin B \cdot \sin C$

2) $\cos(2A) + \cos(2B) + \cos(2C) = 1 - 4\cos A \cdot \cos B \cdot \cos C$

3) $\sin A + \sin B + \sin C = 4\cos(A/2)\cos(B/2)\cos(C/2)$

4) $\cos A + \cos B + \cos C = 1 + 4\sin\left(\frac{A}{2}\right)\sin\left(\frac{B}{2}\right)\sin\left(\frac{C}{2}\right)$

To prove any of the 4 identities above

(i) Club any two terms of LHS and apply sum to product formula

(ii) Apply double angle formula for remaining third term of LHS

(iii) Now take something common and make use of the given condition.

Proof of 1 ⇒

$$LHS = (\sin 2A + \sin 2B) + (\sin 2C)$$
$$= 2\sin(A+B)\cos(A-B) + 2\sin C\cos C$$

$\left[\because A+B+C = \pi \Rightarrow A+B = \pi - C\right.$
$\left.\sin(A+B) = \sin C\right]$

$$= 2\sin C \cdot \cos(A-B) + 2\sin C \cdot \cos C$$
$$= 2\sin C\left[\cos(A-B) + \cos C\right]$$

$\left[\because A+B = \pi - C \Rightarrow \cos(A+B) = -\cos C\right]$

$$= 2\sin C\left[\cos(A-B) - \cos(A+B)\right]$$
$$= 2\sin C\,(2\sin A \cdot \sin B)$$
$$= 4\sin A \cdot \sin B \cdot \sin C = R.H.S$$

5) $\tan A + \tan B + \tan C = \tan A \cdot \tan B \cdot \tan C$

6) $\tan\left(\frac{A}{2}\right)\tan\left(\frac{B}{2}\right) + \tan\left(\frac{B}{2}\right)\tan\left(\frac{C}{2}\right) + \tan\left(\frac{A}{2}\right)\tan\left(\frac{C}{2}\right) = 1$

7) $\cot(A)\cot(B) + \cot(B)\cot(C) + \cot(A)\cot(C) = 1$

8) $\cot\left(\frac{A}{2}\right) + \cot\left(\frac{B}{2}\right) + \cot\left(\frac{C}{2}\right) = \cot\left(\frac{A}{2}\right)\cot\left(\frac{B}{2}\right)\cot\left(\frac{C}{2}\right)$

9) $\sin(A)\cos(B)\cos(C) + \cos(A)\cdot\sin(B)\cos(C) + \sin(C)\cos(B)\cos(A) = \sin A \cdot \sin B \cdot \sin C$

10) $\sin(A)\sin(B)\cos(C) + \sin(A)\cos(B)\sin(C) + \cos(A)\sin(B)\sin(C) = 1 + \cos A \cdot \cos B \cdot \cos C$